

NETWORKING FOUNDATIONS

WEEK1

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1. OSI MODEL

- The **OSI (Open Systems Interconnection)** model is a conceptual model developed by the **International Organization for Standardization (ISO)** that describes how communications should occur in a computer network.
- In other words, the OSI model defines a framework for computer network communications.
- The OSI model is composed of **seven layers**:
 1. Physical Layer
 2. Data Link Layer
 3. Network Layer
 4. Transport Layer
 5. Session Layer
 6. Presentation Layer
 7. Application Layer

Layer 1

- Layer 1: Physical Layer
- The physical layer, also referred to as layer 1, deals with the physical connection between devices; this includes the medium, such as a wire, and the definition of the binary digits 0 and 1.
- Data transmission can be via an electrical, optical, or wireless signal. Consequently, we need data cables or antennas, depending on our physical medium.

Layer2

The data link layer focuses on sending data between two nodes on the same network segment

- Layer 2, known as the Data Link Layer, is responsible for sending data between devices on the same local network. It helps devices communicate with each other by using unique addresses called MAC addresses.
- The data link layer describes an agreement between the different systems on the same network segment on how to communicate.
- A network segment refers to a group of networked devices using a shared medium or channel for information transfer.
- MAC stands for Media Access Control. They are usually expressed in hexadecimal format with a colon separating each two hexadecimal digits (one byte).

Layer 3

- The network layer [layer 3] is concerned with sending data between different networks. In more technical terms, the network layer handles logical addressing and routing, i.e., finding a path to transfer the network packets between the diverse networks.
- Examples of the network layer include Internet Protocol (IP), Internet Control Message Protocol (ICMP), and Virtual Private Network (VPN) protocols such as IPSec and SSL/TLS VPN.

Layer 4

- Layer 4, the transport layer, enables end-to-end communication between running applications on different hosts
- Examples of layer 4 are Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).

LAYER 5 session layer

- The session layer is responsible for **establishing, maintaining, and synchronising communication between applications running** on different hosts.
- Establishing a session means initiating communication between applications and negotiating the necessary parameters for the session.
- Data synchronisation ensures that data **is transmitted in the correct order and** provides mechanisms for recovery in case of transmission failures.
- Examples of the session layer are Network File System (NFS) and Remote Procedure Call (RPC).

layer 6: Presentation Layer

- The presentation layer ensures the data is delivered in a form the application layer can understand.
- Layer 6 **handles data encoding, compression, and encryption.**
- An example of encoding is character encoding, such as ASCII or Unicode.

Layer 7: Application Layer

- the application layer provides network services directly to end-user applications.
- The application layer is the top layer, and you might have encountered many of its protocols as you use different applications. Examples of Layer 7 protocols are HTTP, FTP, DNS, POP3, SMTP, and IMAP

OSI MODEL

Layer Number	Layer Name	Main Function	Example Protocols and Standards
Layer 7	Application layer	Providing services and interfaces to applications	HTTP, FTP, DNS, POP3, SMTP, IMAP
Layer 6	Presentation layer	Data encoding, encryption, and compression	Unicode, MIME, JPEG, PNG, MPEG
Layer 5	Session layer	Establishing, maintaining, and synchronising sessions	NFS, RPC
Layer 4	Transport layer	End-to-end communication and data segmentation	UDP, TCP
Layer 3	Network layer	Logical addressing and routing between networks	IP, ICMP, IPSec
Layer 2	Data link layer	Reliable data transfer between adjacent nodes	Ethernet (802.3), WiFi (802.11)
Layer 1	Physical layer	Physical data transmission media	Electrical, optical, and wireless signals

TCP/IP MODEL

- **TCP/IP stands for Transmission Control Protocol/Internet Protocol**
- It was developed in the **1970s by the Department of Defense (DoD)**.
- from top to bottom, we have:
 - 4. Application Layer:** The OSI model application, presentation and session layers, i.e., layers 5, 6, and 7, are grouped into the application layer in the TCP/IP model.
 - 3. Transport Layer:** This is layer 4.
 - 2. Internet Layer:** This is layer 3. The OSI model's network layer is called the Internet layer in the TCP/IP model.
 - 1. Link Layer:** This is layer 2.

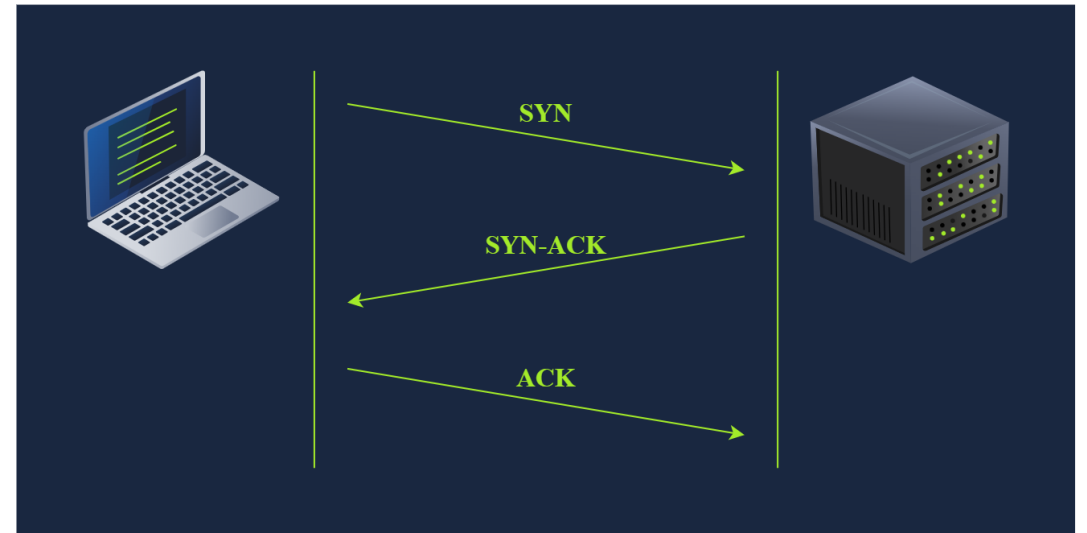
- In the TCP/IP model, we go from top to bottom. The layers are structured starting with the Application Layer at the top, followed by the Transport Layer, Internet Layer, and finally the Link Layer at the bottom. This means the Application Layer is the first layer you interact with, and the Link Layer is the last layer that interfaces with the actual network hardware

TCP/IP model

Layer Number	ISO OSI Model	TCP/IP Model (RFC 1122)	Protocols
7	Application Layer	Application Layer	HTTP, HTTPS, FTP, POP3, SMTP, IMAP, Telnet, SSH,
6	Presentation Layer	Transport Layer	TCP, UDP
5	Session Layer		
4	Transport Layer		
3	Network Layer	Internet Layer	IP, ICMP, IPSec
2	Data Link Layer	Link Layer	Ethernet 802.3, WiFi 802.11
1	Physical Layer		

Three-way handshake

- A TCP connection is established using what's called a three-way handshake.
- Two flags are used: **SYN (Synchronise)** and **ACK (Acknowledgment)**. The packets are sent as follows:
- **SYN Packet:** The client initiates the connection by sending a SYN packet to the server. This packet contains the client's randomly chosen initial sequence number.
- **SYN-ACK Packet:** The server responds to the SYN packet with a SYN-ACK packet, which adds the initial sequence number randomly chosen by the server.
- **ACK Packet:** The three-way handshake is completed as the client sends an ACK packet to acknowledge the reception of the SYN-ACK packet.
- a valid port number ranges between 1 and 65535 because it uses two octets and port 0 is reserved.



TCP flags

TCP Flags — The Language of Every Scan

SYN

"I want to connect"

ACK

"I received your packet"

RST

"Close immediately"

FIN

"Done, close gracefully"

PSH

"Deliver data now"

URG

"Process immediately"

NULL

No flags — stealth



Mnemonic: SYN = Says Yes Now · ACK = Acknowledged · RST = Reject Suddenly · FIN = Finished

TCP

- **TCP (Transmission Control Protocol)** is a **connection-oriented** transport protocol. Being connection-oriented, it **requires the establishment** of a TCP connection before any data can be sent.
- It ensure **reliable data delivery sent** by the different processes on the networked hosts.
- In TCP, **each data octet has a sequence number**; this makes it easy for the receiver to identify lost or duplicated packets. The receiver, on the other hand, acknowledges the reception of data with an acknowledgement number specifying the last received octet.

Each TCP connection uses two ports:

Source Port (from the sender)

Destination Port (to the receiver)

Common TCP Ports & Their Uses:

Port Number	Service	Description
80	HTTP	Web browsing (insecure)
443	HTTPS	Secure web browsing
22	SSH	Secure remote login
21	FTP	File Transfer Protocol
25	SMTP	Sending emails
110	POP3	Receiving emails
143	IMAP	Email retrieval
3306	MySQL	Database access
3389	RDP	Remote Desktop Protocol

How TCP Ports Work

- 1. Client Sends a Request** → The client device selects a random source port and connects to the server's destination port (e.g., port 443 for HTTPS).
- 2. TCP Handshake** → TCP establishes a reliable connection using a three-way handshake (SYN, SYN-ACK, ACK).
- 3. Data Transfer** → The server and client exchange data , ensuring no loss.
- 4. Connection Termination** → Once data transfer is complete, the connection is closed properly.

UDP

- **UDP (User Datagram Protocol)** is a **connectionless protocol** that operates at the transport layer, i.e., layer 4.
- there is **no guarantee that the UDP packet has been received successfully.**
- It **better speed** than a transport protocol that provides “confirmation.”

How UDP Works?

- ➡ **Client Sends a Datagram** → A UDP message (datagram) is sent to the destination port without establishing a connection.
- ➡ **No Acknowledgment** → The receiver gets the data, but there's no confirmation if it was received correctly.
- ➡ **Data Processing** → If a packet is lost or arrives out of order, UDP does not attempt to fix it (unlike TCP).

Common UDP Ports & Their Uses

Port Number	Service	Description
53	DNS	Resolves domain names to IPs
67/68	DHCP	Assigns IP addresses dynamically
123	NTP	Network Time Protocol (syncs clocks)
161/162	SNMP	Network monitoring & management
500	IPsec	VPN security protocol
69	TFTP	Simple file transfers
1900	SSDP	Device discovery on a network

DHCP → Dynamic Host Configuration Protocol

NTP → Network Time Protocol

SNMP → Simple Network Management Protocol

IPsec → Internet Protocol Security

TFTP → Trivial File Transfer Protocol

SSDP → Simple Service Discovery Protocol

Protocols : -

A protocol is a set of **rules and standards** that define how data transmitted and received between devices in a network or system.

Types of Protocol

- **Networking Protocols:**

- Used for communication between devices over the internet or local networks.
- **Examples:**
 - **TCP (Transmission Control Protocol)** → Reliable, ordered data transmission.
 - **UDP (User Datagram Protocol)** → Fast, connectionless communication.
 - **IP (Internet Protocol)** → Defines how devices are addressed using IP addresses.
 - **HTTP (Hyper Text Transfer Protocol)** → Used for web browsing.
 - **HTTPS (Secure HTTP)** → Secure web communication using encryption.
 - **FTP (File Transfer Protocol)** → Transfers files between computers.
 - **NAT(Network Address Translation)** – translates private ip into public ip address.

Types of protocol :-

- **Security Protocols**

- Used to secure communication and protect data.

- Examples:

1. **SSL/TLS (Secure Sockets Layer / Transport Layer Security)** → Encrypts web traffic (HTTPS).
2. **IPSec (Internet Protocol Security)** → Secures VPN connections.
3. **SSH (Secure Shell)** → Secure remote login.
4. **Kerberos:** Ticket-based authentication protocol for internal enterprise services

- **3. Email Protocols**

- Used for sending and receiving emails.

- Examples:

1. **SMTP (Simple Mail Transfer Protocol)** → Sends emails.
2. **POP3 (Post Office Protocol v3)** → Downloads emails from a server.
3. **IMAP (Internet Message Access Protocol)** → Accesses emails on a server.

Types of protocol :-

- Wireless & IoT Protocols

- Used for **wireless communication and Internet of Things (IoT)** devices.

- **Examples:**

- 1. **Wi-Fi** → Wireless networking.

- 2. **Bluetooth** → Short-range communication.

- 3. **MQTT (Message Queuing Telemetry Transport)** → Lightweight IoT messaging.

ports

❏ Port

- Ports are virtual or physical connection points used for data transfer between computers.

1. Physical Ports

- Physical ports are hardware-based connection points found on computers. Examples include USB, Ethernet, HDMI, and DisplayPort, which are used to connect various devices to the computer and facilitate data transfer.

2. Virtual Ports

- Virtual ports are numbered between 0 and 65535 and are used for communication over a network. Each port number is associated with specific services or protocols, and all data transfer over the internet occurs through these virtual ports.
- For example, when visiting websites, we connect to port 80 or 443 on the server hosting the website.

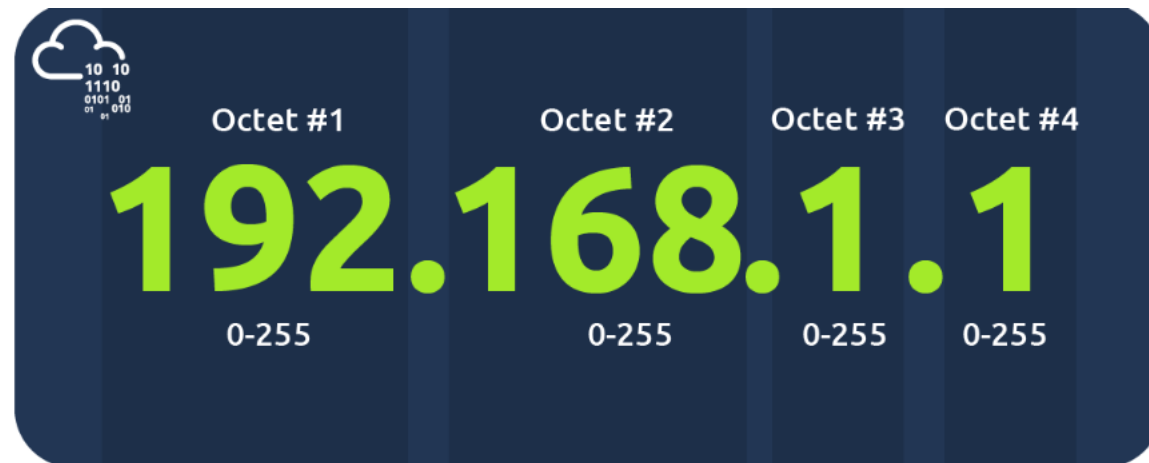
ports

➤ Commonly used services and protocols operate on specific ports. Below is a table of some frequently used ports and services:

• Port Number	Service / Protocol
21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
80	HTTP
443	HTTPS
3306	MySQL
3389	RDP
5432	PostgreSQL

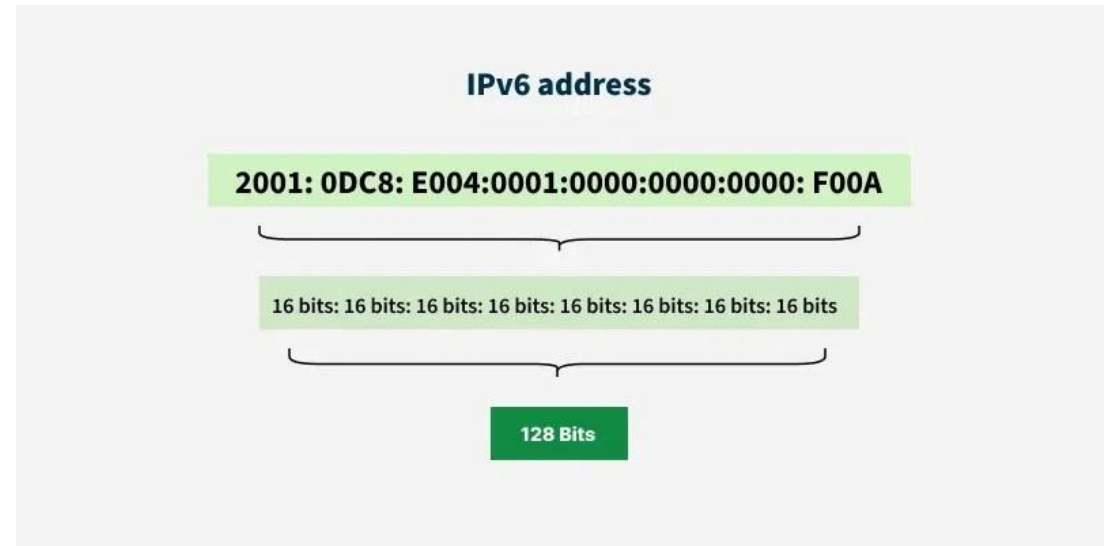
Ip address

- An IP address comprises four octets, i.e., 32 bits. Being 8 bits, an octet allows us to represent a decimal number between 0 and 255.
- Every host on the network needs a unique identifier for other hosts to communicate with him. Without a unique identifier, the host cannot be found without ambiguity. When using the TCP/IP protocol suite, we need to assign an IP address for each device connected to the network.



IPv6:

- IPv6: Developed due to the insufficient addressing capacity of IPv4, it uses a 128-bit addressing system



mac

- Devices have the same thing: two means of identification, with one being permeable.
- A Media Access Control (MAC) Address -- think of this as being similar to a serial number.